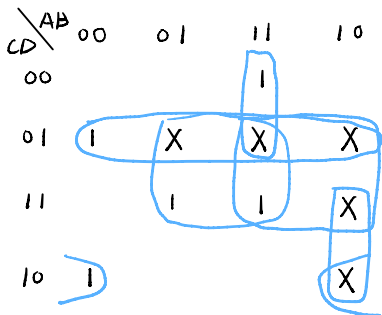


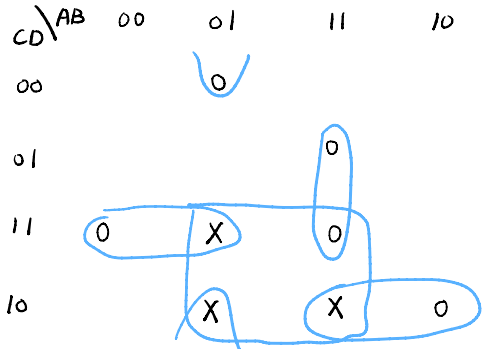
1. 解: ① 卡诺图:



② 蕴含项: $C'D, BD, AD, ABC', AB'C, B'CD'$
实质蕴含项: $C'D, BD, ABC', B'CD'$

③ 化简结果: $f(A, B, C, D) = C'D + BD + ABC' + B'CD'$

(b) ① 卡诺图



② 蕴含项: $B'C', A+B'+D, A+C'+D', A'+B'+D', A'+C'+D, A'+B'+D', A'+C'+D$
实质蕴含项: $A+B'+D, A+C'+D, A'+B'+D', A'+C'+D$

③ 化简结果: $f(A, B, C, D) = (A+B'+D)(A+C'+D')(A'+B'+D')(A'+C'+D)$

2. 解: ① 最小项表

0	0000	✓	
1	0001	✓	0, 1 000 - ✓ 0, 1, 8, 9 - 00 - PI_1
4	0100	✓	0, 4 0-00 ✓ 0, 4, 8, 12 -- 00 PI_2
8	1000	✓	0, 8 -000 ✓
9	1001	✓	1, 9 -001 ✓
12	1100	✓	4, 12 -100 ✓
			8, 9 100 - ✓
7	0111	✓	8, 12 1-00 ✓
11	1011	✓	9, 11 10-1 PI_3
15	1111	✓	7, 15 -111 PI_4
			12, 15 1-11 PI_5

② 蕴含项表

	0	1	4	7	9	12
✓ PI_1	x	⊗			x	
✓ PI_2	x		⊗			⊗
PI_3					x	
✓ PI_4				⊗		
PI_5						

$\Rightarrow f(A, B, C, D) = PI_1 + PI_2 + PI_4 = B'C' + C'D' + BCD$

3. 解: ① 划分法

P_0	A	B	C	D	E	F	G	H
$x=0$	1	0	0	0	0	1	0	1
$x=1$	0	0	0	0	0	0	0	0
P_1	(A F H)	(B C D E G)						
$x=0$	A	B	H	C	B	E	D	D
$x=1$	E	F	G	F	H	F	A	H
P_2	(A H)	(F)	(B C D E G)					
$x=0$	A	H	B	C	B	E	D	D
$x=1$	E	G	F	F	H	F	A	H
P_3	(A H)	(F)	(B D)	(C E G)				
$x=0$	A	H	B	C	E	B	D	D
$x=1$	E	G	F	F	F	H	A	H
P_4	(A H)	(F)	(B D)	(C E G)				

$\Rightarrow$	$A, H \rightarrow (A H)$	present	next	output
			$x=0$	$x=1$
	$F \rightarrow (F)$	(A H)	(A H)	(C E G)
	$B, D \rightarrow (B D)$	(F)	(B D)	(F)
	$C, E, G \rightarrow (C E G)$	(B D)	(C E G)	(F)
		(C E G)	(B D)	(A H)

② 蕴含表法

B	x						
C	x	FH					
D	x	CE	BE				
E	x	CF	BF	AD			
F	x	AF	BF	AD	AF		
G	x	CF	BF	AD	AF	X	
H	x	CF	BF	AD	AF	X	X

$A, H \rightarrow (A H)$	present	next	output
		$x=0$	$x=1$
$B, D \rightarrow (B D)$	(A H)	(A H)	(C E G)
$C, E, G \rightarrow (C E G)$	(F)	(B D)	(F)
$F \rightarrow (F)$	(B D)	(C E G)	(F)
	(C E G)	(B D)	(A H)

4. 解: ① 规则1: 对一个输入具有相同次态的状态, 编码相邻.

状态对: $[(F), (C E G)], [(F), (B D)]$

② 规则2: 一个现态在逻辑相邻输入下的次态, 由状态编码相邻.

状态对: $[(A H), (C E G)], [(B D), (F)],$

$[(C E G), (F)], [(B D), (A H)].$